



Can Fragmentation in Scaffold-Constrained Molecular Generation Be Solved with Algebraic Graph Connectivity Scoring?

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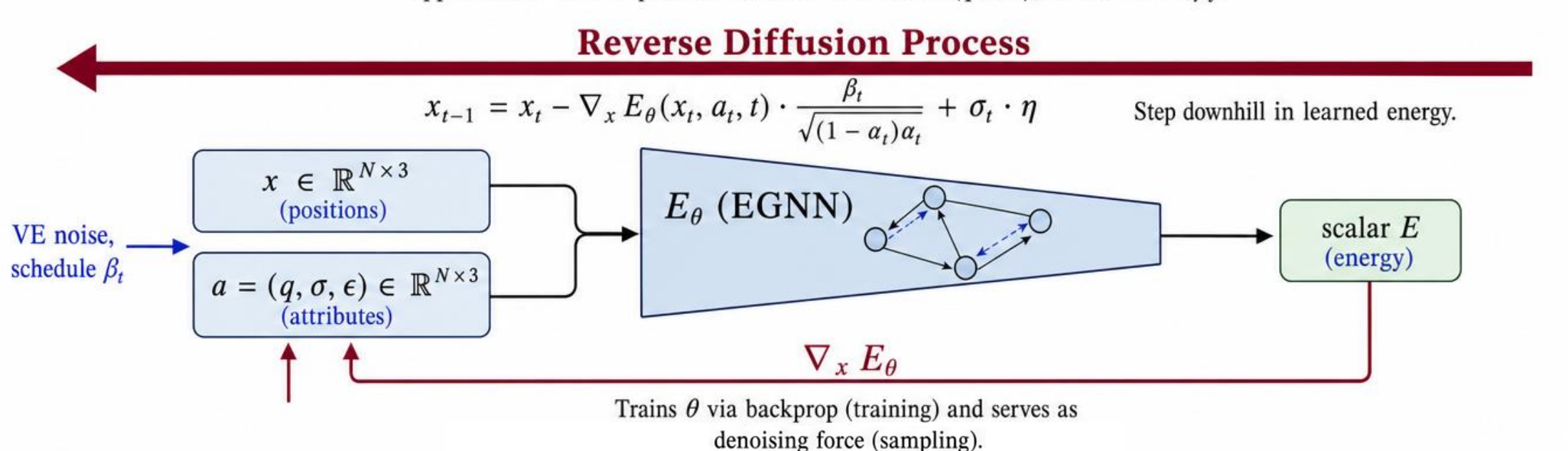
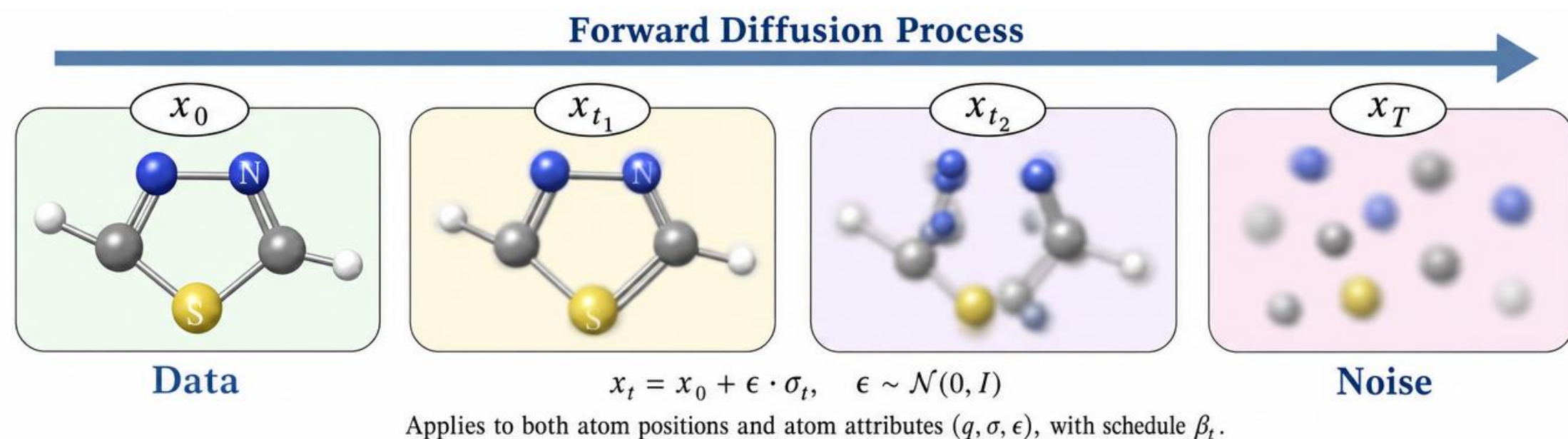
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Motivation & Background

- Existing **de novo molecular generation** models, like our **LEVI** (Learned Energy Via Iterative Denoising) model, produce candidates from pure noise and use processes like **diffusion** in order to procedurally noise up spatial and chemical attributes and learn to denoise them to **generate new conformations**.



- However, existing de novo models are trained to generate **entire molecules**. In R&D, if a **scaffold** is already known to bind a target, like a fragment hit or a piece sitting in a **binding pocket**, chemists want to grow it into a **full drug candidate** without disturbing that binding, so those atoms have to stay **fixed** while the rest is generated around them. This is called **molecular inpainting**.

De novo models are not trained for inpainting, and even if they are finetuned, existing diffusion models (inpainting or not) produce **disconnected molecules**.

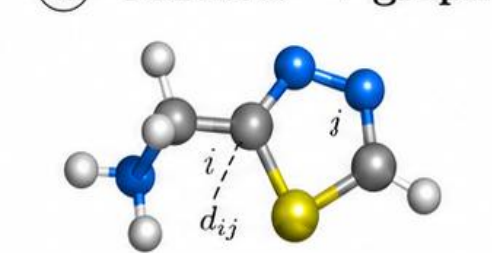
- Existing literature has exploited molecular structure as **graphs** for **graph neural network** use cases, but not to consider molecular connectedness as **algebraic connectivity**.

Research Objectives

- Implement an algebraic connectivity force (**ACForce**) in order to improve generated molecules' connectedness.
- Improve connectedness in both **pretrained de novo model (LEVI)** and **finetuned inpainting diffusion models**, across **different anchor connection modes**.
- Retain **respect for valences** and **existing chemistry** in generated molecules.

Algebraic connectivity is based on the **Fiedler's eigenvalue maximization problem** of the **graph Laplacian's** adjacency matrix.

① Contacts $\rightarrow$ graph



Distances between atoms define a smooth weighted adjacency matrix A_t (the edge weights). The nodes are the residues (atoms). All results that follow use the Laplacian matrix L constructed from A .

③ Fiedler value (λ_2)

Order the eigenvalues of L : $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_m$

$\lambda_2 > 0 \iff G$ is connected

λ_2 (the second-smallest eigenvalue, or Fiedler value) measures algebraic connectivity: higher λ_2 means the graph is more robustly connected.

Variational characterization:

$$\lambda_2 = \min_{x \perp \mathbf{1}, x \neq 0} \frac{x^T L x}{x^T x}$$

$x \perp \mathbf{1}$ means the trivial constant solution ($x^T \mathbf{1} = 0$).

Numerator $x^T L x$ = total weighted disagreement (roughness) of x across edges.

Denominator $x^T x$ normalizes the scale of x .

② Graph Laplacian

Encode the graph's connectivity:

$$D = \text{diag}(A\mathbf{1}) \quad (D_{ii} = \sum_j A_{ij}) \quad (D: \text{degree matrix})$$
$$L = D - A \quad (L: \text{combinatorial Laplacian})$$
$$L = L^T \geq 0, \quad L\mathbf{1} = 0$$

($\mathbf{1}$: all-ones vector; $\mathbf{0}$: all-zeros vector)

The Laplacian is the matrix encoding local connectivity. Every subsequent result follows from this matrix.

Here, L is symmetric positive semidefinite and has a zero eigenvalue with eigenvector $\mathbf{1}$.

④ ACForce loss function definition

Since $L\mathbf{1} = 0$, the smallest eigenvalue is always $\lambda_1 = 0$. If there is more than one connected component, there is another independent zero eigenvector.

$$\lambda_2 = 0 \iff \text{another zero eigenvector exists} \iff \text{graph is disconnected}$$
$$\lambda_2 > 0 \iff \text{graph is connected}$$

Define the loss as the negative Fiedler value:

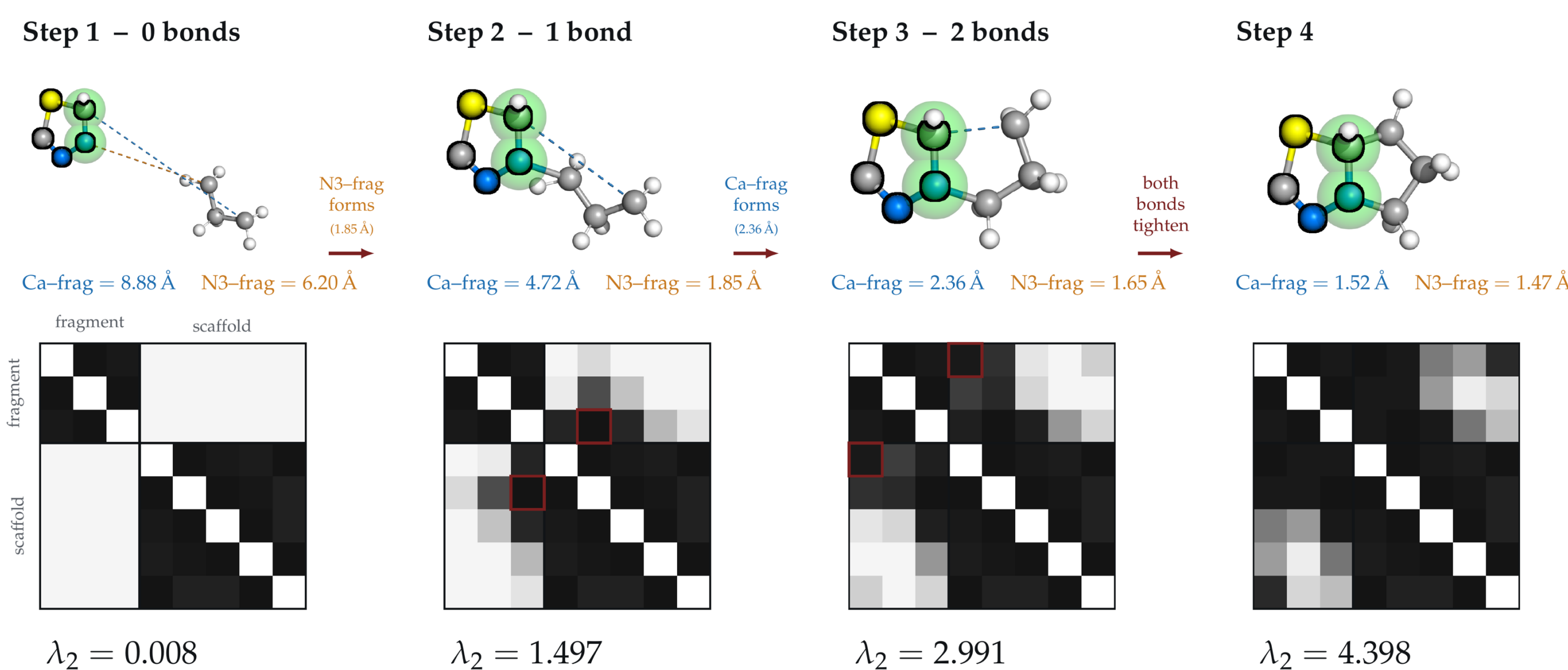
$$\mathcal{L}(r) = -\lambda_2(L(r)) \quad (\text{where } L(r) \text{ is the Laplacian of the soft adjacency from positions } r)$$

Compute forces (negative gradient of $\mathcal{L}$ w.r.t. atomic coordinates r):

$$F(r) = -\nabla_r \mathcal{L} = \nabla_r \lambda_2(L(r))$$

via autograd through the eigendecomposition of $L(r)$.

Forces are the gradient of λ_2 w.r.t. atomic coordinates r . They push atoms to increase λ_2 , i.e., to make the graph more robustly connected.

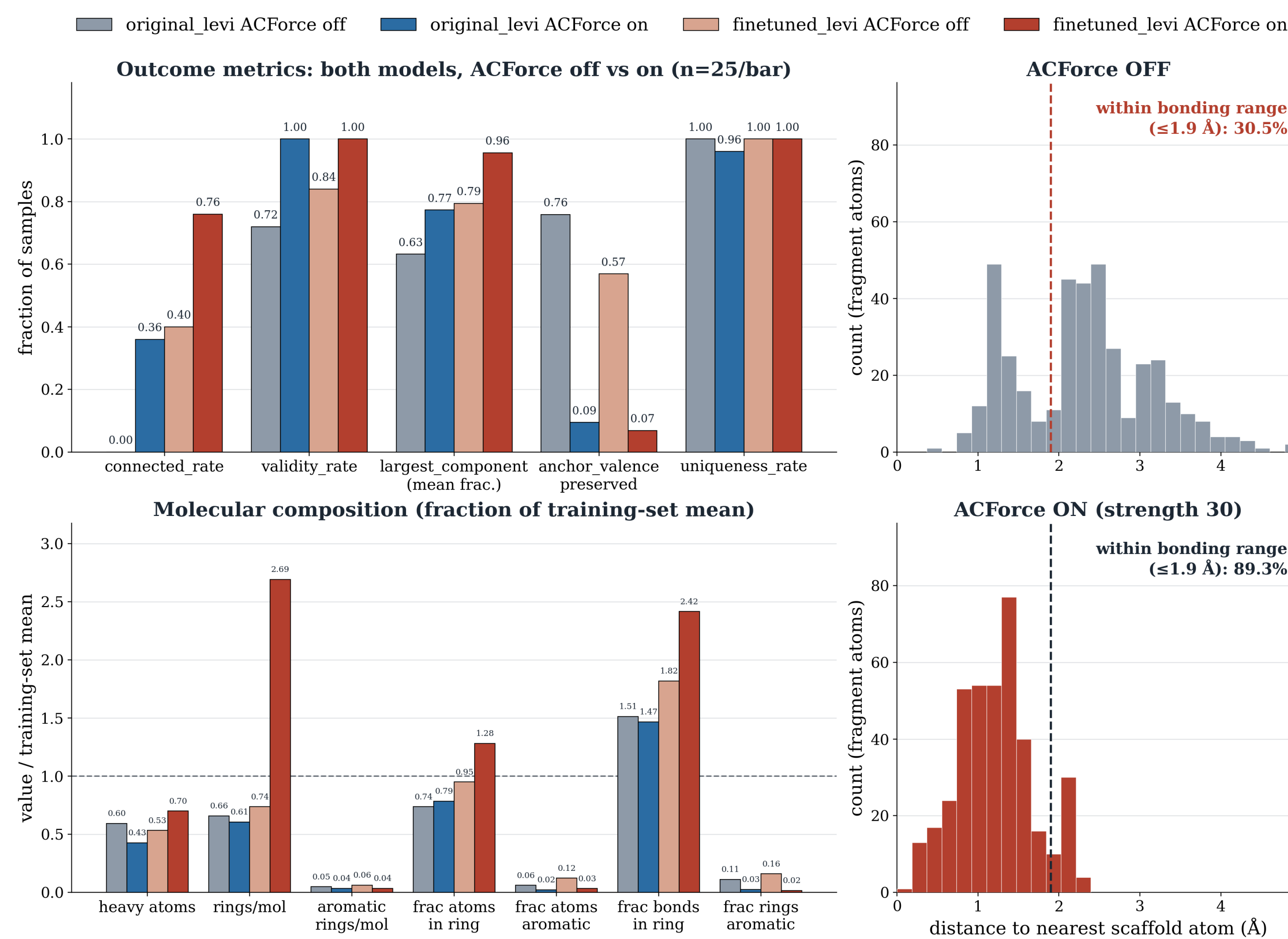


cell shade $\propto A_{ij} = \text{sig}((r_c - d_{ij})/\tau)$

far $A_{ij} \rightarrow 0$ close $A_{ij} \rightarrow 1$

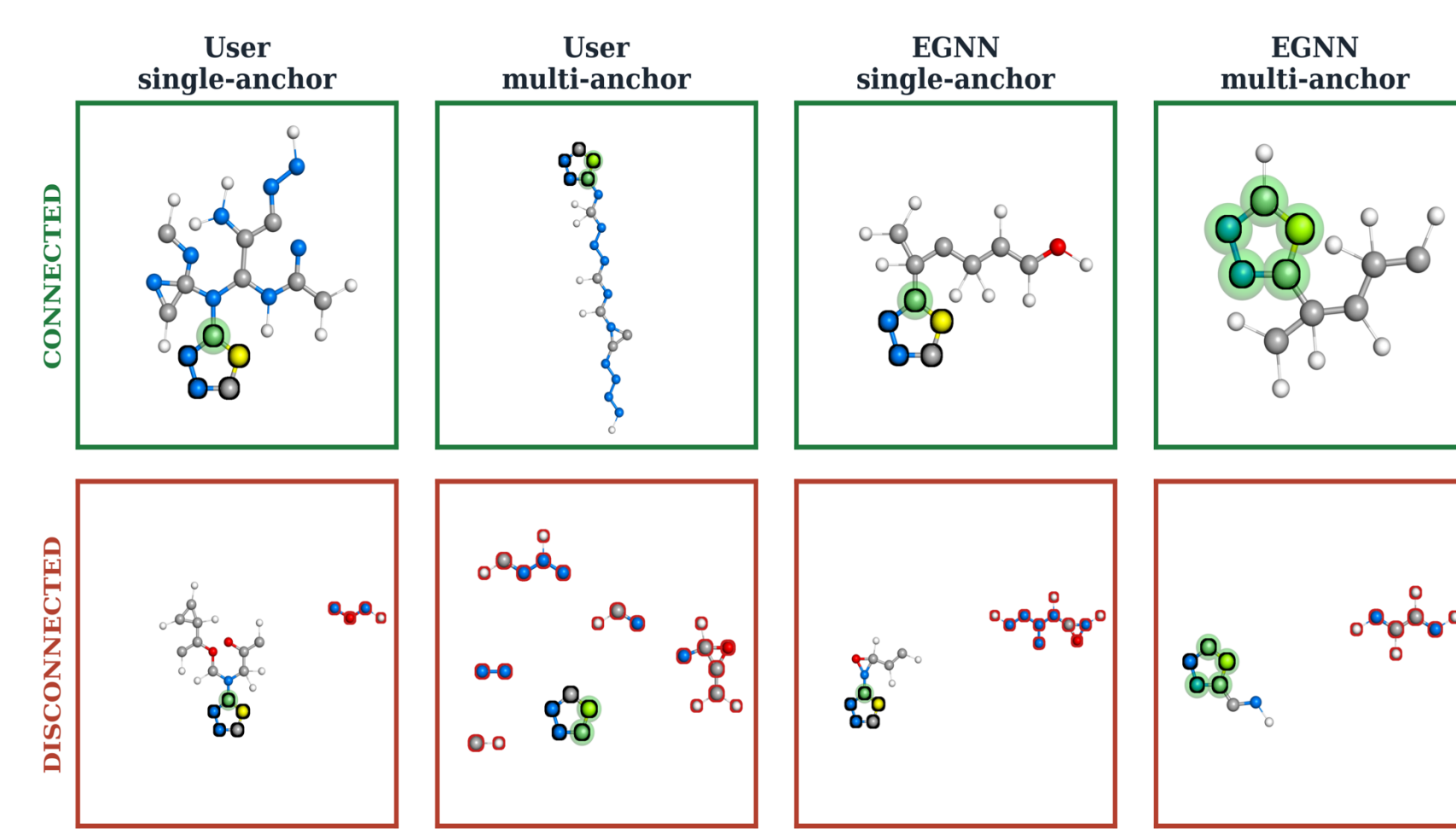
boxed cell = the scaffold-fragment entry that just crossed into a bond this step

Using ACForce on the best model improved connectedness from **40% to 76%**.



Benchmarking the **finetuned** inpainting model on **1,3,4-thiadiazole** as a scaffold, connectedness **increases significantly in 3 of 4 anchor modes** to as much as **72%**.

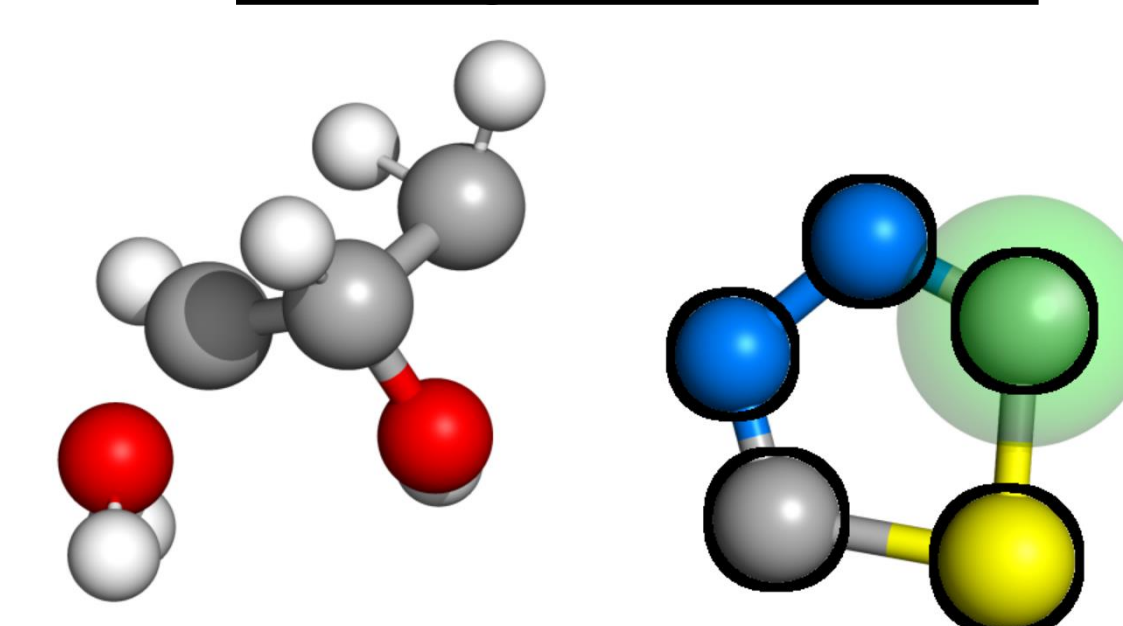
Selected Connected & Disconnected Samples across Anchor Modes with Finetuned ACForce Model



Conclusions & Limitations

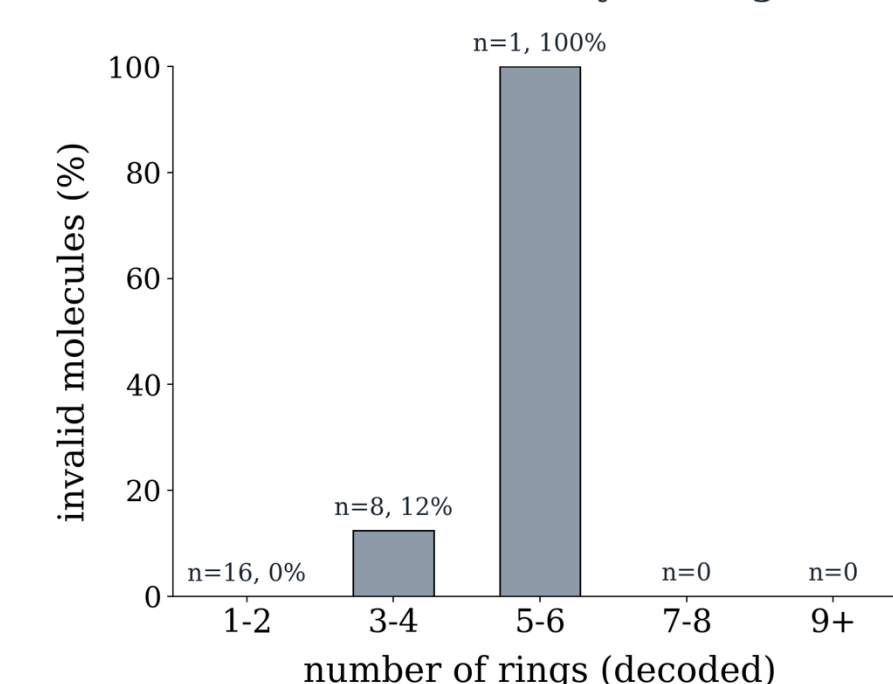
- Significant connectivity improvement in **3 of 4** anchor-selection modes:
- user-single-anchor: **p = 0.02**
- user-multi-anchor: **p < 0.0001**
- EGNN-multi-anchor: **p < 0.0001**
- NOT in EGNN-single mode: p=0.20, conn. 60% $\rightarrow$ 51%
- ACForce produces **valid valences** and **chemistry** according to RDKit!

EGNN Single-Anchor Prediction



Fragment generates on opposite side as single anchor and can never connect

ACForce off: invalidity vs. ring count



C12H3N3OS

Overgeneration of Rings and Ring Frequency Mismatch of Finetuned Model with Training Set

Next Steps

- Implement ACForce to determine if it can fix fragmentation within the **base LEVI model**, so that de novo generated molecules are all in one piece. Test different **r_c values in the soft adjacency mapping**.
- Benchmark ACForce and LEVI against **DiffLinker's non-energy-based method** with the **GEOM dataset** for connectedness and validity.
- Determine **chemical motif overrepresentation** and mitigation strategies to **improve generative diversity**.
- Run a case study for inpainting on a **known drug target** in a **protein binding pocket** to gauge applicability and potential failure modes, including **single-anchor inpainting**.
- Develop a **learnable threshold activation function** such that **λ_2 grows from [0,1]** and then **tapers off past 1** to avoid **over-compactness** of molecules that **reduces generative diversity and robustness**, e.g. $f(\lambda_2; L, k) = L - (L - 1)e^{-k(\lambda_2 - 1)}$. This can also be done with an intrafragment submatrix loss to encourage fragments to connect to themselves before overbonding with a scaffold.

Selected References & Links

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Diffusion



Connectivity

Scan the QR codes to view **explainer videos** about diffusion modelling and Fiedler's algebraic connectivity of the graph Laplacian!

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